

Potential features / proposed scope

September 10, 2026 / P01 through P08. Status labels apply to the feature, not the entire future product.

Feature / status	Capability and boundary
P01 / Standalone engineering application Proposed	Own workspace, native project representation, feature/assembly behavior, execution runtime and deployed model route for supported workflows. Scope: Demonstrate useful work without launching Fusion or relying on an external AI service.
P02 / Shared manual and agent editing Proposed	Use the same commands for direct manipulation and conversational edits, preserving editable history and design intent. Scope: Show a second human edit can rebuild correctly after agent work.
P03 / Selection, drawing, image and voice context Proposed	Attach requests to actual faces/features and coordinate frames; accept drawings, annotated images or voice context alongside exact references. Scope: Resolve selection and units explicitly; do not infer hidden dimensions as known facts.
P04 / Engineering conversation and teaching Proposed	Explain concepts, critique designs, explore mechanism alternatives and ask for consequential missing inputs while independent work can continue. Scope: Evaluate useful clarification and checked explanations on representative tasks.
P05 / Unified workspace views Proposed	Switch among 3D, assembly/feature trees, parameters, drawings, analysis, manufacturing, BOM, requirements and history without losing task context. Scope: Build actual two-way connections to a versioned project model.
P06 / Requirements and acceptance contracts Proposed	Track user requirements, assumptions, units, tolerances, materials, interfaces, objectives, priorities and acceptance evidence. Scope: Keep supplied, measured, supplier, calculated and estimated values distinguishable.
P07 / Project branches and alternatives Proposed	Compare revisions and alternative designs on mass, cost, constraints and manufacturing assumptions; merge reviewable change sets. Scope: Implement revision guards and conflict handling, not just named chat histories.
P08 / Editable parametric features Proposed	Add sketch dimensions/constraints, extrude/revolve/loft, holes, patterns, fillets and dependency-aware feature regeneration. Scope: Prove rebuild and editability on unseen in-scope designs.

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Potential features / proposed scope

September 10, 2026 / P09 through P15. Status labels apply to the feature, not the entire future product.

Feature / status	Capability and boundary
P09 / Assembly relationships and motion Proposed	Represent parts/configurations/transforms/joints and assess degrees of freedom, interference and clearance across declared motion. Scope: Validate assembly constraints and sampled motion; model checks do not establish real friction.
P10 / Stable geometry references Proposed	Use feature-origin identity, semantic roles and geometric matching to preserve references through topology changes. Scope: Fail visibly when remapping is ambiguous; identifiers alone do not solve topology.
P11 / Native geometry and constraints Proposed	Operate a precise boundary-representation kernel and constraint solvers with derived display meshes. Scope: Kernel licensing/development, feature behavior and persistence require separate validation.
P12 / Import, exchange and repair Proposed	Report imported units, hierarchy, metadata, repairs and missing history; support geometry exchange and explicitly inferred feature recovery. Scope: Quantify translation losses. Proprietary translators and native formats remain feasibility work.
P13 / Mesh, scan and reconstruction workflows Proposed	Allow mesh/scan data and image-based design interpretation alongside exact CAD, with visible precision and editability differences. Scope: Treat reconstructed hidden geometry and unknown scale as inference, not measured truth.
P14 / Engineering drawings and BOMs Proposed	Generate revision-linked views, dimensions, datums, tolerances, notes, quantities and configurations with source-dated purchased-part data. Scope: Validate drawings/BOM completeness and their input revision.
P15 / Analytical and numerical simulation Proposed	Connect equations, meshing and solver workers for supported structural/FEA, thermal, fluid/CFD or other declared analysis classes. Scope: Each class needs validated loads, restraints, material models, convergence checks and benchmarks; coverage is not selected.

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Potential features / proposed scope

September 10, 2026 / P16 through P22. Status labels apply to the feature, not the entire future product.

Feature / status	Capability and boundary
P16 / Manufacturing and CAM preparation Proposed	Plan stock, tools/holders, fixtures, setups, operation sequences and postprocessed toolpaths; check process-specific limits and tool reach. Scope: Qualify each process/machine route separately. Toolpath generation does not authorize machine execution.
P17 / Evidence-linked verification Proposed	Connect requirements to data/unit checks, CAD integrity, assembly checks, analytical/numerical results, manufacturing review and physical tests. Scope: Show methods, scope and limitations instead of one generic confidence score.
P18 / Change impact and stale-result detection Proposed	Trace which edits invalidate downstream drawings, analyses, BOM entries, purchased interfaces and toolpaths. Scope: Recompute affected evidence or demonstrate that it is unaffected.
P19 / Physical test feedback Proposed	Attach procedures, instruments, calibration and measured results to the as-built configuration; compare actual outcomes with predictions. Scope: Digital model success remains separate from physical qualification.
P20 / Reusable engineering libraries Proposed	Store approved components, interfaces, material references and organizational practices with source/revision metadata. Scope: Reusing a part does not automatically reuse the validity of an old load case.
P21 / Persistent task execution Proposed	Checkpoint the action-observation loop with explicit task states, artifact references, bounded retries and task budgets. Scope: Recover from process restart and disconnected UI without blind mutation replay.
P22 / Independent background workers Proposed	Separate the UI from CAD, solver, rendering and inference workers so a viewer can disconnect while an authorized host keeps running. Scope: A sleeping local host cannot compute; remote continuation needs an actual authorized worker.

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Potential features / proposed scope

September 10, 2026 / P23 through P29. Status labels apply to the feature, not the entire future product.

Feature / status	Capability and boundary
P23 / Coordinated multi-agent work Proposed	Assign scoped design, analysis, manufacturing and documentation roles; serialize dependent writes and branch independent alternatives. Scope: Multiple agents are not automatically independent verification or unlimited parallel CAD.
P24 / Revision guards and operation receipts Proposed	Check target identity, units, permissions, preconditions and expected revision outside the model; persist mutation outcomes. Scope: Reconcile uncertain external changes after failures before retrying.
P25 / Recovery, undo and manual takeover Proposed	Preserve partial progress, diagnose failures, separate cancel from undo and hand the editing lease back to the user. Scope: Demonstrate recovery with interruption, stale-target and user-edit tests.
P26 / Local, hybrid and hosted deployment Proposed	Offer consistent project/tool contracts across local-only, remote-assisted and organization-hosted configurations. Scope: All selected models/workers must be available in local-only mode; no deployment mode is shipped independently yet.
P27 / Resource and task budgets Proposed	Measure tokens, solver time, CPU/GPU time and storage where available, with task/day budgets, priorities and concurrency controls. Scope: Validate end-to-end resource use; a timer or token total alone is not a money cap.
P28 / Production delivery and support Proposed	Provide tested cross-platform installers/updates, migrations, backups, authentication, telemetry, rollback and incident/support procedures. Scope: Public licensing, support commitments, final branding and commercial pricing remain undecided.
P29 / Independent local or hosted inference Proposed	Deploy a suitably licensed model on an owned inference route with replaceable provider adapters. Scope: Benchmark candidate quality, latency, memory and total cost per accepted task.

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Potential features / proposed scope

September 10, 2026 / P30 through P36. Status labels apply to the feature, not the entire future product.

Feature / status	Capability and boundary
P30 / Multimodal engineering model Proposed	Interpret language, diagrams and rendered geometry, query exact project facts, select typed operations and repair tool failures. Scope: No proprietary engineering model has been trained; model size and architecture are not selected.
P31 / Geometry-aware model representations Long-term option	Explore encoders for feature graphs/geometric descriptors and hierarchical retrieval over large assemblies. Scope: Show measured value over simpler structured tools and retrieval.
P32 / Revision-matched visual grounding Proposed	Combine reproducible perspectives/sections/exploded views/analysis overlays with entity ID maps and exact geometric queries. Scope: Tie every image and measurement to the same revision.
P33 / Project memory and reference retrieval Proposed	Keep exact project state, conversation context, cited reference documents and learned weights distinct; retrieve relevant objects and sources. Scope: Respect project access and flag stale documents or supplier information.
P34 / Permissioned trajectory datasets Proposed	Curate expert, consenting-user, licensed and generated tasks with starting state, actions, failures, corrections and checked outcomes. Scope: Require source rights, evidence, deduplication and project/family-separated evaluation data.
P35 / Evaluated model specialization Proposed	Compare supervised or parameter-efficient adaptation, broader fine-tuning, preferences and constraint-aware reward learning where appropriate. Scope: Training rewards must reflect valid geometry and satisfied constraints; promotion depends on unseen-task results.
P36 / Versioned model releases Proposed	Maintain candidate checkpoints, evaluation reports, gradual deployment, regression monitoring and rollback. Scope: Updating project memory is not immediate retraining of general model skill.

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Potential features / proposed scope

September 10, 2026 / P37 through P43. Status labels apply to the feature, not the entire future product.

Feature / status	Capability and boundary
P37 / Specialized model roles Long-term option	Optionally route planning, perception and repair through a family of models while presenting one engineering assistant. Scope: Use extra specialization only if evaluation justifies its complexity.
P38 / Foundation model from scratch Long-term option	Consider original large-scale model training as a later research direction. Scope: Requires sufficient data, compute, capital and demonstrated benefit; not required for the first independent release.
P39 / Organization data controls Proposed	Add explicit training consent, project exclusions, contributor rights, export/deletion controls, secret filtering and private component libraries. Scope: Access to a design is not permission to train; complete governance is not supplied by the current exporter.
P40 / Enforced action and data boundaries Proposed	Apply project read/write, export, code, remote upload, compute, purchasing and hardware permissions in software, with protected credentials. Scope: Imported instructions cannot grant permissions; unrestricted code requires an actual containment strategy.
P41 / Independent held-out benchmarks Proposed	Test requirements, geometry, editability, clarification, recovery, wrong-target changes, latency, usage and reviewer correction time on unseen design families. Scope: Evaluator-controlled inputs/oracles must remain outside the agent workspace; no final held-out suite exists yet.
P42 / Real-user workflow validation Proposed	Measure useful repeated outcomes against a complete manual baseline, including setup, corrections and review. Scope: A mounting-plate change order is a candidate benchmark, not a selected market or validated customer result.
P43 / Progressively broader supported scope Proposed	Expand assemblies, drawing systems, simulation, manufacturing processes and large-model performance only with new evidence. Scope: The long-term breadth associated with mature CAD suites is an ambition, not current parity or a delivery promise.

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